

Linear and non-linear relationships

UNIT 6: Quadratic graphs

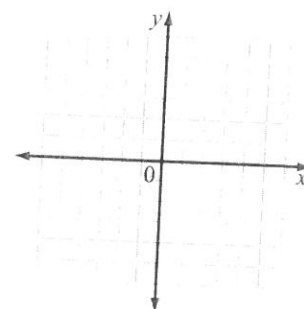
QUESTION 1 Complete the table of values and then, on the same number plane, draw the graphs of the following.

a $y = x^2$

b $y = 2x^2$

c $y = \frac{1}{2}x^2$

x	-3	-2	-1	0	1	2	3
$y = x^2$							
$y = 2x^2$							
$y = \frac{1}{2}x^2$							



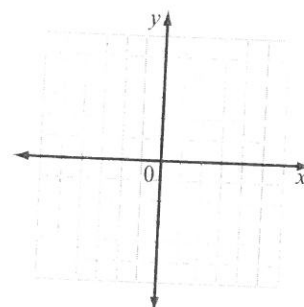
QUESTION 2 Complete the table of values and then, on the same number plane, draw the graphs of the following.

a $y = x^2$

b $y = x^2 + 1$

c $y = x^2 - 1$

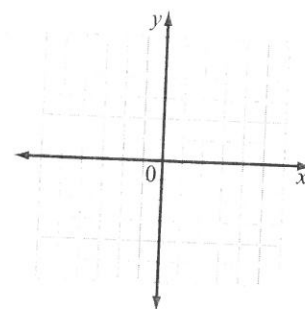
x	-3	-2	-1	0	1	2	3
$y = x^2$							
$y = x^2 + 1$							
$y = x^2 - 1$							



QUESTION 3 Complete the table of values for $y = 1 - x^2$ and sketch its graph.

x	-3	-2	-1	0	1	2	3
$1 - x^2$							

- What is the equation of its axis of symmetry? _____
- What are the coordinates of its vertex? _____
- What is the maximum value for $y = 1 - x^2$? _____
- Find the x -intercepts. _____
- Find the y -intercept. _____



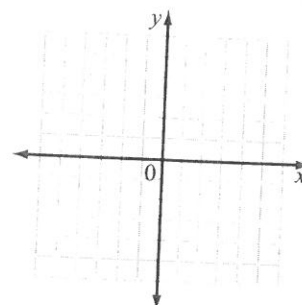
QUESTION 4 Sketch the graphs of the following.

a $y = x^2$

b $y = x^2 + 2$

c $y = x^2 - 2$

- d Explain how the graphs of $y = x^2 + 2$ and $y = x^2 - 2$ can be drawn using $y = x^2$.



Linear and non-linear relationships

UNIT 4: Parallel and perpendicular lines (1)

QUESTION 1 Complete.

- a Lines that are parallel have gradients that are _____.
- b Lines that are perpendicular have gradients that are _____.

QUESTION 2 Determine whether the two given lines are parallel or perpendicular or neither.

- | | | | |
|----------------|--------------------------|---------------------------|-----------------------------------|
| a $y = 2x + 3$ | b $y = \frac{1}{2}x + 5$ | c $y = \frac{3}{2}x + 1$ | d $y = \frac{x}{4} + \frac{2}{3}$ |
| $y = 2x - 1$ | $y = 2x - 3$ | $y = -\frac{2}{3}x - 1$ | $y = -4x + 2$ |
| _____ | _____ | _____ | _____ |
| e $y = 6 - x$ | f $y = 1 - 2x$ | g $y = -\frac{3}{4}x - 2$ | h $y = \frac{5x}{6} - 7$ |
| $y = -x + 3$ | $y = \frac{x}{2} + 5$ | $y = -\frac{4}{3}x + 2$ | $y = -\frac{6}{5}x + 1$ |
| _____ | _____ | _____ | _____ |

QUESTION 3 Write down the equation of the line that passes through the point (0, 2) and which is parallel to the given line.

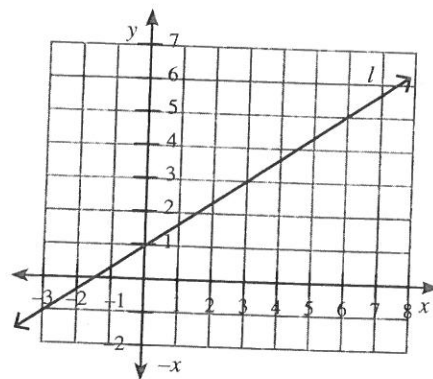
- | | |
|------------------------------------|---------------------------|
| a $y = 3x + 1$ | b $y = -2x - 3$ |
| _____ | _____ |
| c $y = \frac{1}{2}x + \frac{1}{4}$ | d $y = -\frac{5x}{3} - 4$ |
| _____ | _____ |

QUESTION 4 Write down the equation of the line that passes through the origin and which is perpendicular to the given line.

- | | |
|----------------|--------------------------|
| a $y = 4x + 3$ | b $y = -\frac{x}{3} + 7$ |
| _____ | _____ |
| c $y = 9 - 2x$ | d $y = \frac{3}{2}x + 4$ |
| _____ | _____ |

QUESTION 5 The diagram shows the graph of line l .

- a What is the gradient of l ? _____
- b What is the gradient of any line parallel to l ? _____
- c What is the gradient of any line perpendicular to l ? _____
- d Line m has equation $y = -\frac{3}{2}x - 12$. It intersects line l at P . If l meets the y -axis at Q and m meets the y -axis at R . What is the size of $\angle QPR$? _____



Further algebra

UNIT 11: Solving quadratic equations by factorising

QUESTION 1 Solve the following quadratic equations by factorising.

a $x^2 + 5x + 6 = 0$

b $x^2 - 2x - 35 = 0$

c $x^2 - 5x - 6 = 0$

d $x^2 + 7x + 12 = 0$

e $x^2 - 5x + 6 = 0$

f $x^2 + 2x - 48 = 0$

g $x^2 - 8x + 16 = 0$

h $x^2 + 2x - 15 = 0$

i $x^2 + 9x + 20 = 0$

j $x^2 - 8x + 15 = 0$

k $x^2 + 4x - 12 = 0$

l $x^2 - 3x - 10 = 0$

m $x^2 + 11x + 30 = 0$

n $x^2 - 9x + 14 = 0$

o $x^2 + 3x - 28 = 0$

p $x^2 - 2x - 99 = 0$

q $x^2 + 6x + 8 = 0$

r $x^2 + 6x - 7 = 0$

s $x^2 - 6x + 5 = 0$

t $x^2 + 8x + 16 = 0$

u $x^2 - 4x - 60 = 0$

QUESTION 2 Factorise and solve the following quadratic equations.

a $x^2 = 3x + 18$

b $x^2 + 40 = 13x$

c $x^2 + 5x = 36$

d $x^2 = 15x - 54$

e $x^2 - 2x = 24$

f $x^2 = 24 - 5x$

Further algebra

UNIT 9: Quadratic equations in factorised form

QUESTION 1 Solve the following quadratic equations that are already expressed in factorised form.

a $(x-1)(x-2) = 0$

b $(x-2)(x+3) = 0$

c $(x-1)(x-3) = 0$

d $x(x+5) = 0$

e $2x(x-4) = 0$

f $(x-3)(x-7) = 0$

g $(x-3)(x-5) = 0$

h $(x+1)(x-3) = 0$

i $(x+2)(x-4) = 0$

j $(x+3)(x-3) = 0$

k $(x-2)(x+2) = 0$

l $(x-5)(x+5) = 0$

m $(x+1)(x-6) = 0$

n $(x+2)(x+3) = 0$

o $x(x+8) = 0$

QUESTION 2 Solve the following quadratic equations.

a $x(2x-1) = 0$

b $(x+6)(2x-1) = 0$

c $(3x-2)(x+1) = 0$

d $(x-2)(3x-1) = 0$

e $5x(2x-1) = 0$

f $3x(x-2) = 0$

g $(x+3)(3x-1) = 0$

h $4x(2x-5) = 0$

i $-2x(x-1) = 0$

j $(3x+1)x = 0$

k $(x-3)^2 = 0$

l $3x(x-3) = 0$

QUESTION 3 Solve the following equations.

a $(x-4)(x-5) = 0$

b $(x-8)(x+8) = 0$

c $x(x-3) = 0$

d $2x(x-2) = 0$

e $(x-7)(x-9) = 0$

f $(x+1)(x-5) = 0$

g $(2x-1)(x+4) = 0$

h $(2x+3)(2x-3) = 0$

i $(4x+5)(5x-4) = 0$

Right-angled triangles and trigonometry

UNIT 7: Mixed problems

QUESTION 1 In $\triangle ABC$, $\angle C = 90^\circ$, $\angle B = 38^\circ 40'$ and $AB = 14.6$ cm.
Find BC correct to one decimal place.

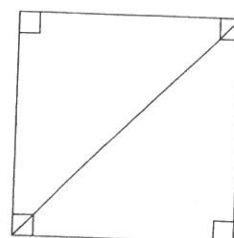
QUESTION 2 A ladder leans against a vertical wall with its foot 1.5 metres from the wall making an angle of $45^\circ 36'$ with the ground. How long is the ladder? Give your answer to the nearest centimetre.

QUESTION 3 A tree 18 metres tall casts a shadow 19.5 metres long.
What angle do the rays of the Sun make with the ground?

QUESTION 4 The height of a ramp is 4.2 m. Given that the ramp is inclined at 30° to the ground, find the length of the ramp to the nearest centimetre.

QUESTION 5 A tree casts a shadow 20.7 m long. If the Sun's rays meet the ground at $29^\circ 36'$, what is the height of the tree to the nearest metre?

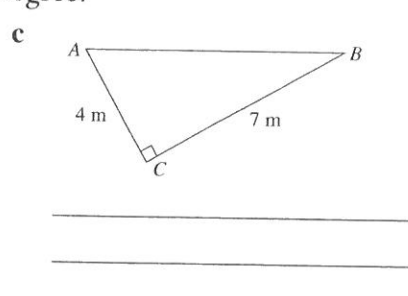
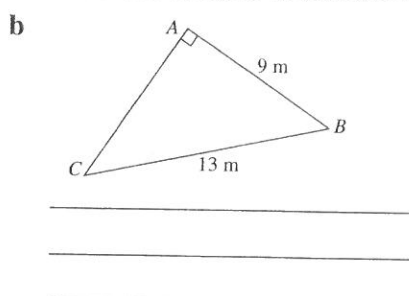
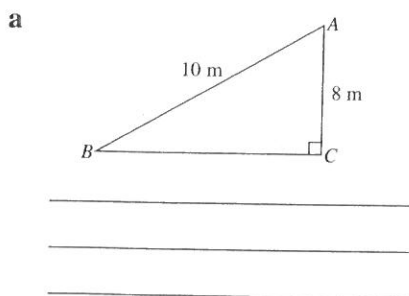
QUESTION 6 The diagonal of a square is 24.6 cm long.
Find the length of one side to the nearest millimetre.



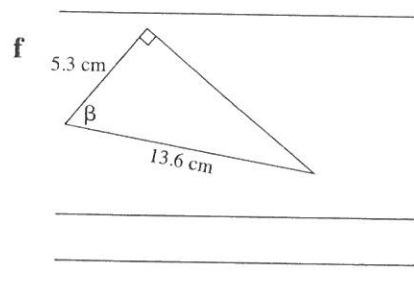
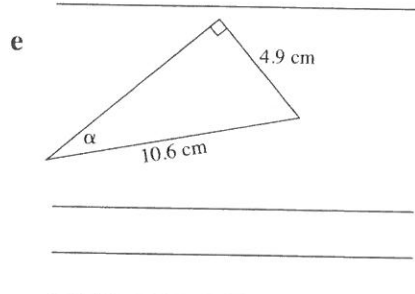
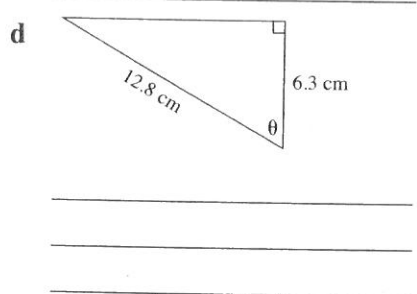
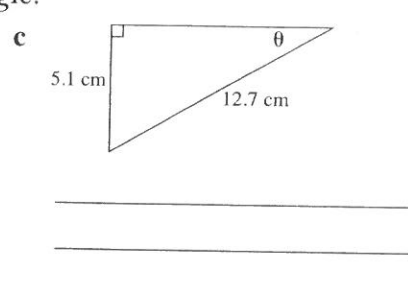
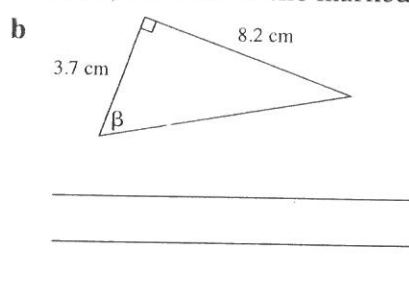
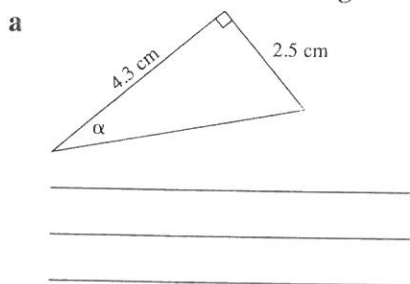
Right-angled triangles and trigonometry

UNIT 6: Finding an unknown angle

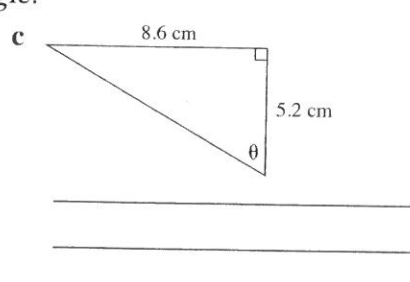
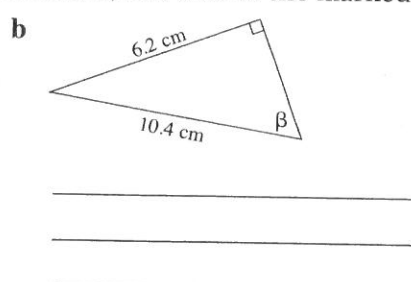
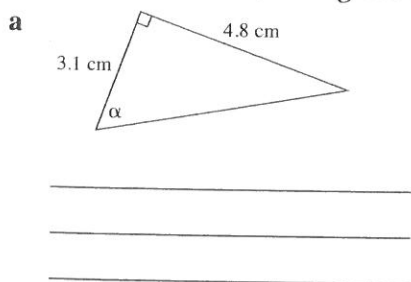
QUESTION 1 Find the size of angle B. Give the answer to the nearest degree.



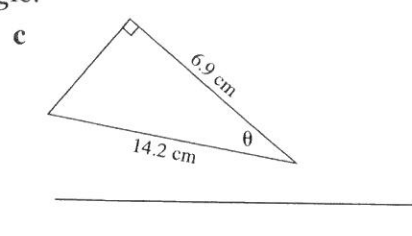
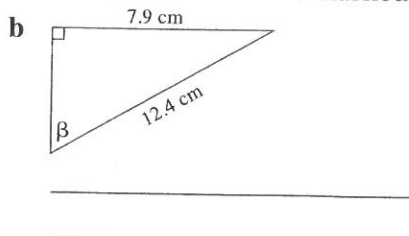
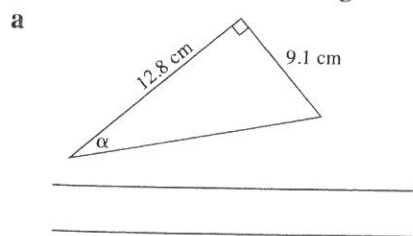
QUESTION 2 Find, in degrees and minutes, the size of the marked angle.



QUESTION 3 Find, in degrees and minutes, the size of the marked angle.



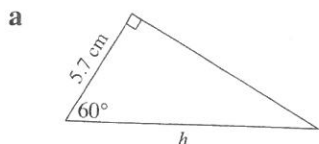
QUESTION 4 Find, in degrees and minutes, the size of the marked angle.

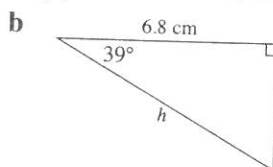


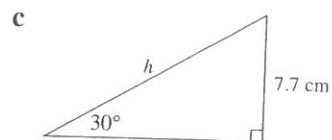
Right-angled triangles and trigonometry

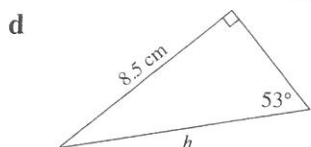
UNIT 5: Finding the hypotenuse

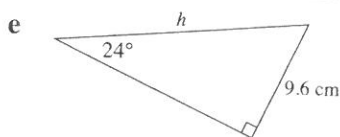
QUESTION 1 Find the length of the hypotenuse correct to two decimal places.

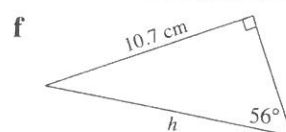




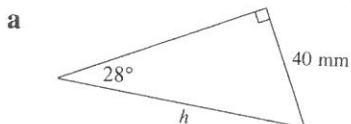


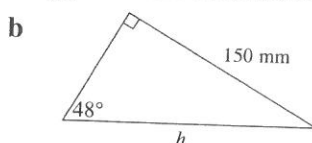


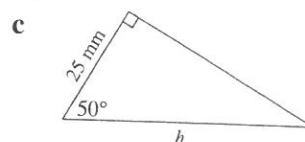


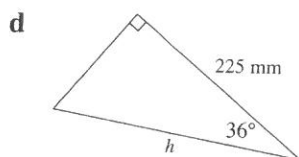


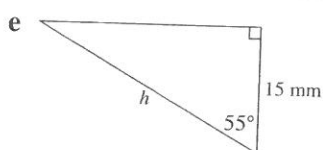
QUESTION 2 Find the length of the hypotenuse correct to one decimal place.

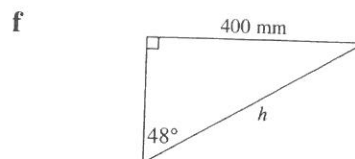




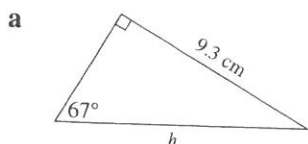


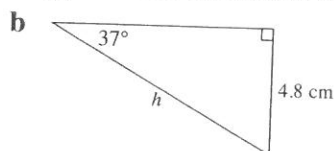


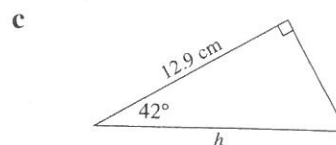




QUESTION 3 Find the length of the hypotenuse correct to two decimal places.



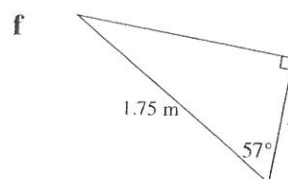
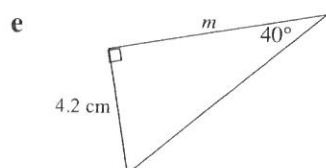
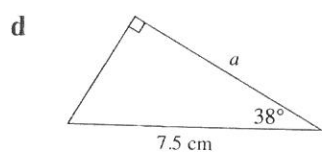
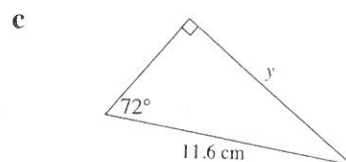
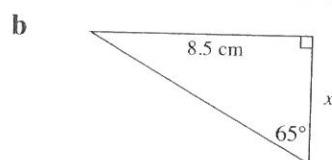
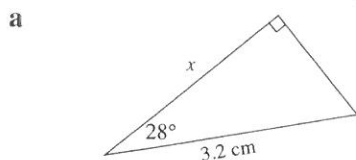




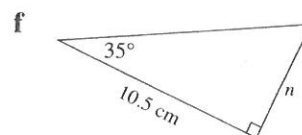
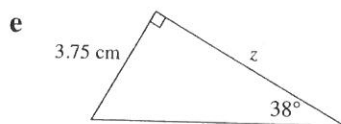
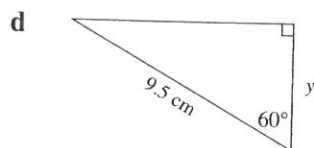
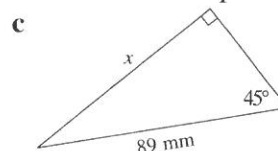
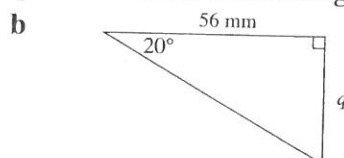
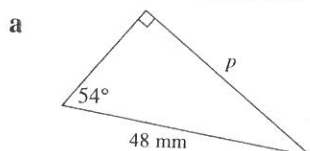
Right-angled triangles and trigonometry

UNIT 4: Finding a side

QUESTION 1 Find the length of the unknown side correct to one decimal place.



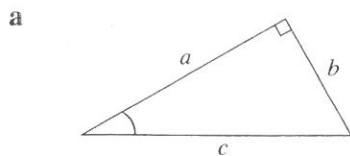
QUESTION 2 Find the value of the pronumeral in each triangle correct to two decimal places.



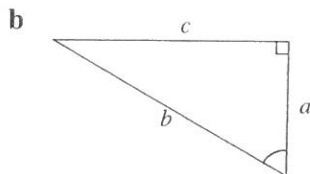
Right-angled triangles and trigonometry

UNIT 2: The trigonometric ratios

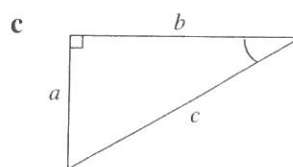
QUESTION 1 In each of the following triangles, state whether a , b and c are the opposite side, adjacent side or hypotenuse with reference to the angle marked.



$a = \underline{\hspace{1cm}}, b = \underline{\hspace{1cm}}, c = \underline{\hspace{1cm}}$

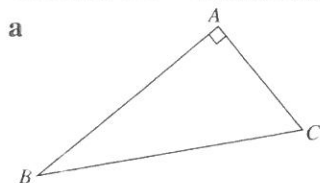


$a = \underline{\hspace{1cm}}, b = \underline{\hspace{1cm}}, c = \underline{\hspace{1cm}}$

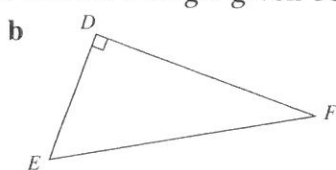


$a = \underline{\hspace{1cm}}, b = \underline{\hspace{1cm}}, c = \underline{\hspace{1cm}}$

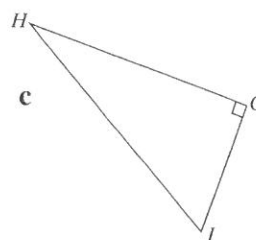
QUESTION 2 Name the hypotenuse in each triangle given below.



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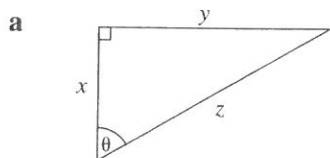


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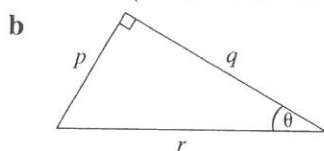


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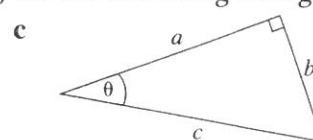
QUESTION 3 Write the trigonometric ratios (sine, cosine and tangent) for the following triangles.



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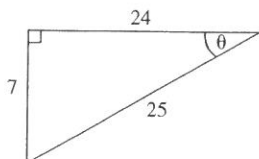


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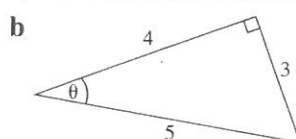


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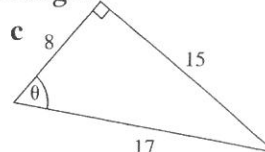
QUESTION 4 Find the fraction $\sin \theta$, $\cos \theta$ and $\tan \theta$ in the following triangles.



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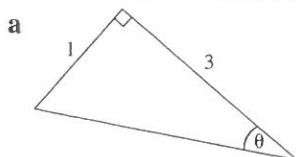


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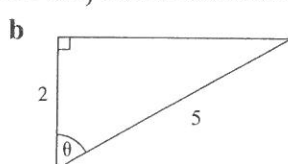


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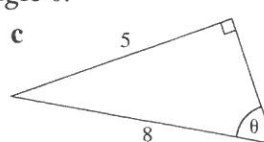
QUESTION 5 Which ratio (sin, cos or tan) could be used to find the angle θ .



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Simultaneous equations

UNIT 8: Mixed questions

QUESTION 1 Solve the following pairs of simultaneous equations by the method of elimination.

a $9a - 2b = 36$
 $a + 2b = 14$

b $2x + y = 8$
 $5x + 2y = -3$

c $x + 3y = 8$
 $3x - y = 9$

QUESTION 2 Use the method of substitution to solve the following pairs of simultaneous equations.

a $2x - y = 9$
 $3x - 2y = 15$

b $3x + 8y = 4$
 $3x + 2y = -2$

c $5m - 2n = 20$
 $3m - 4n = 12$

QUESTION 3 Solve the following simultaneous equations by any suitable method.

a $x + y = 10$
 $3x - 4y = 9$

b $x + 5y = 15$
 $-x + 2y = 6$

c $2x + y = -8$
 $3x - 2y = -12$

d $3x + y = 7$
 $x + 2y = 9$

e $3x + 5y = 25$
 $2x - y = 8$

f $2x - 5y = 13$
 $5x - 3y = -15$

Simultaneous equations

UNIT 7: The method of elimination

QUESTION 1 Solve the following pairs of simultaneous equations by the method of elimination.

a $x + 2y = 8$
 $3x - 2y = 4$

b $2x - 3y = 6$
 $x + 3y = 9$

c $2x + 5y = 19$
 $3x - 5y = 6$

d $2x + y = 10$
 $3x - y = 5$

e $3x + 4y = 14$
 $x + y = 3$

f $2x - y = 3$
 $x - 2y = 9$

QUESTION 2 Use the method of elimination to solve the following pairs of simultaneous equations.

a $3x + 4y = 12$
 $x - y = 4$

b $4x + 5y = 22$
 $x + y = 10$

c $2x + 3y = 11$
 $x - y = -2$

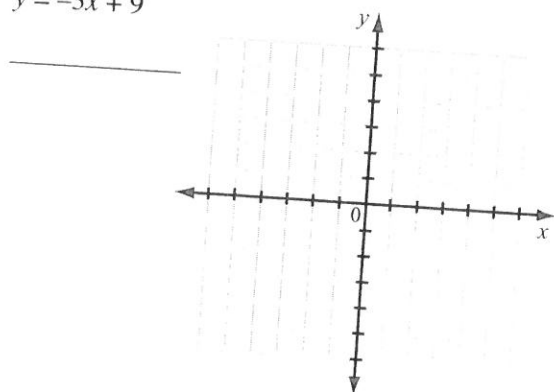
d $5x - 3y = 9$
 $3x + y = 4$

Simultaneous equations

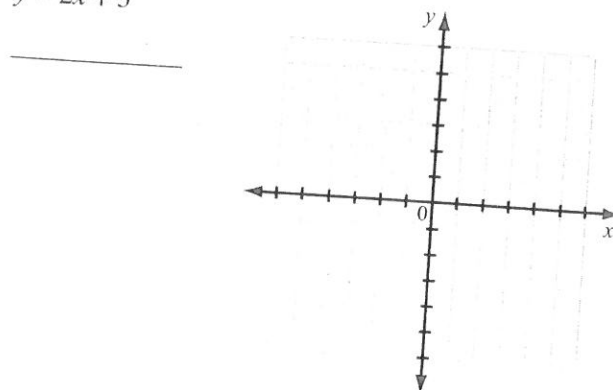
UNIT 3: The graphical method

QUESTION 1 Solve, by drawing graphs, the following pairs of simultaneous equations.

a $y = x + 5$
 $y = -3x + 9$

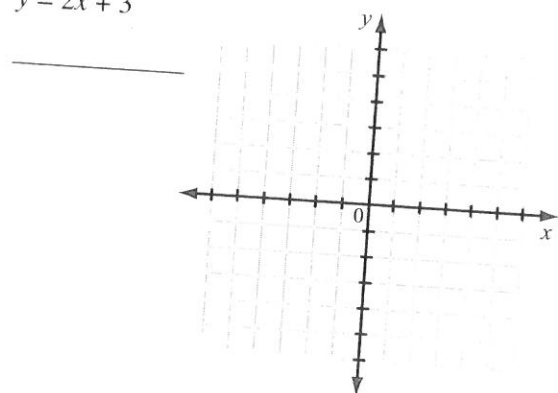


b $y = x + 3$
 $y = 2x + 5$

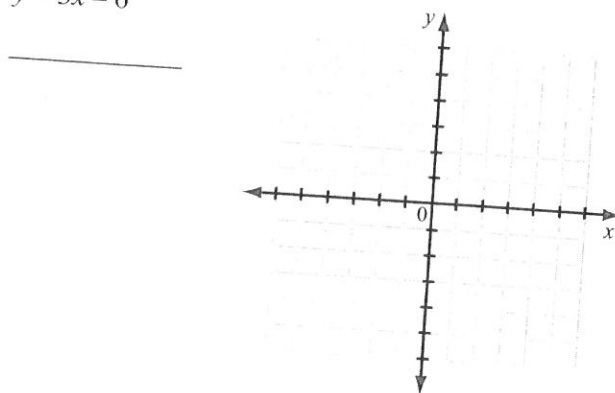


QUESTION 2 Solve graphically the following pairs of simultaneous equations.

a $y = x + 1$
 $y = 2x + 3$

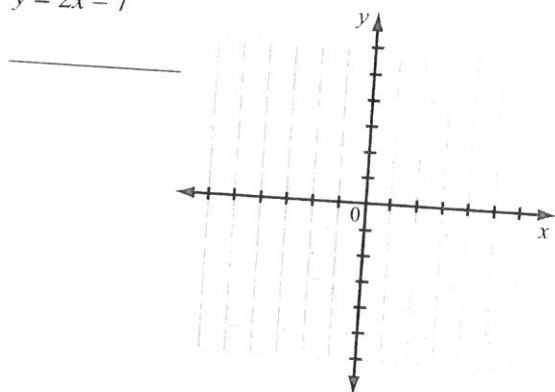


b $y = x - 4$
 $y = 3x - 6$



QUESTION 3 Graph each pair of equations on the same number plane to find their solution.

a $y = -x + 2$
 $y = 2x - 7$



b $x + y = 3$
 $2x - y = -9$

